

Matrix inequalities and norms

Minimum number of orthogonal conjugations for diagonal pinching

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: An exact all-dimension optimum imposes rigid real orthogonality constraints beyond a general upper bound; it would clarify matrix averaging and majorization decompositions.

Partial result — 2026-09-11

Partial result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS for the stated partial result.**

The fixed and adaptive orthogonal pinching lengths both equal the least row count $h(d)$ of a sign matrix H with $H^T H = h(d)I_d$. In particular, the exact length is 12 for $9 \leq d \leq 12$.

Still open: The general value of $h(d)$ is undetermined; the all-dimension optimization remains open and includes Hadamard-order existence questions. The ratings apply to this surviving question.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Lean proof and verification evidence — 2026-09-14

The stated fixed-list partial result is Lean verified; the canonical problem remains Partially resolved. The fixed-list feasibility characterization by rectangular sign matrices is proved for every positive dimension and row count. The rank and divisibility obstructions and an explicit order-twelve sign matrix establish that the actual minimum fixed pinching length is 12 for every dimension from 9 through 12. The source's adaptive comparison is not formalized, and the all-dimension optimum remains open.

Immutable formal proof. The checked exports are `NLA.MI08.fixed_sign_equivalence`, `NLA.MI08.design_obstructions`, `NLA.MI08.hadamard_twelve`, `NLA.MI08.finite_minimums`. Lean 4.33.1 uses pinned Mathlib `0df444a3` and LeanCert `621a43d7`; [complete pins and reproduction instructions](#) are retained. All four exported transitive axiom closures contain only `propext`, `Classical.choice` and `Quot.sound`.

Formalization: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, with OpenAI Codex assistance. Matthew J. Colbrook retains mathematical authorship. Two independent statement reviews preceded implementation, and two independent final proof reviews passed. Kernel-only LeanCert and [fresh isolated Linux Comparator and kernel verification](#) passed, including rejection controls. [Original artifact](#), [exact checked inputs](#) and [independent operational audit](#) are retained. This is AI-assisted formal verification, not external human peer review or source-author endorsement.

Problem statement

For each integer $d \geq 1$, define $\Delta(X) = \text{diag}(x_{11}, \dots, x_{dd})$ for $X = (x_{ij}) \in \mathbb{R}^{d \times d}$. Determine the integer

$$\varphi(d) = \min \left\{ q \geq 1 : \exists U_1, \dots, U_q \in O(d) \forall X \in \mathbb{R}^{d \times d}, \quad \Delta(X) = \frac{1}{q} \sum_{i=1}^q U_i X U_i^T \right\},$$

where $O(d) = \{U \in \mathbb{R}^{d \times d} : U^T U = I_d\}$. The same list of orthogonal matrices must work for every X . The minimum is finite: $q = 2^{\lceil \log_2 d \rceil}$ is always possible.

Why it matters

Diagonal extraction is a basic matrix operation. Expressing it by the shortest average of orthogonal similarities gives an exact decomposition complexity and sharpens real versions of majorization-based norm estimates.

References

1. J.-C. Bourin and E.-Y. Lee, *Averages over matrix unitary orbits and spectral order*, arXiv:2606.15624v2 (18 June 2026), Lemma 4.1 and Question 4.8. [Primary text](#).
2. R. Bhatia, *Pinching, trimming, truncating, and averaging of matrices*, American Mathematical Monthly 107(7) (2000), 602–608, equation (2), for the complex unitary counterpart cited in reference 1. [DOI](#).

Status check — 2026-09-10

The latest source is v2 and explicitly asks for the smallest length. Searches included *orthogonal pinching minimum phi Bourin Lee, Averages over matrix unitary orbits spectral order 2026*, and *diagonal pinching orthogonal matrices minimum*. No general determination was located. The adjacent Question 4.9 reverses the quantifier order by allowing the matrices to depend on X ; it is not counted separately here. This is an exact optimization question rather than a conjectured closed formula.

Audit update (2026-09-10): Rechecked Question 4.8 in the current Bourin–Lee text and searched for minimal orthogonal pinching averages. The question remains explicit; allowing the conjugations to depend on X is the different Question 4.9. This is a bounded literature check, not a proof that no solution exists.